Chapter 5: Numbers in the Vedic Literature

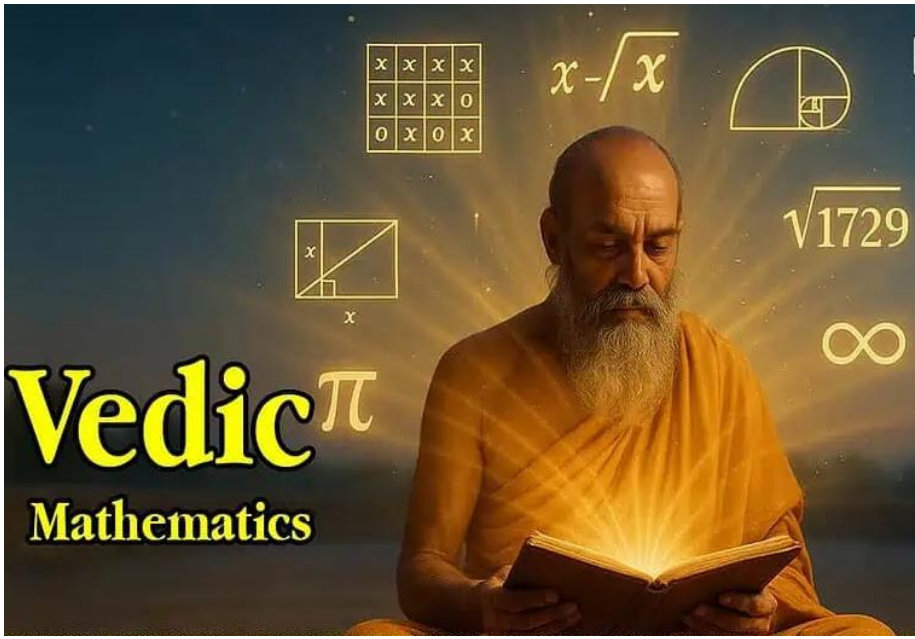
The Mathematics of India by P. P. Divakaran

Introduction to Vedic Numbers

- Vedic literature contains some of the earliest references to numbers in India.

- Numbers were deeply connected to ritual, cosmology, and philosophy.

- Counting and enumeration were essential for Vedic sacrifices.



- Large numbers appear frequently in hymns and ritual texts.

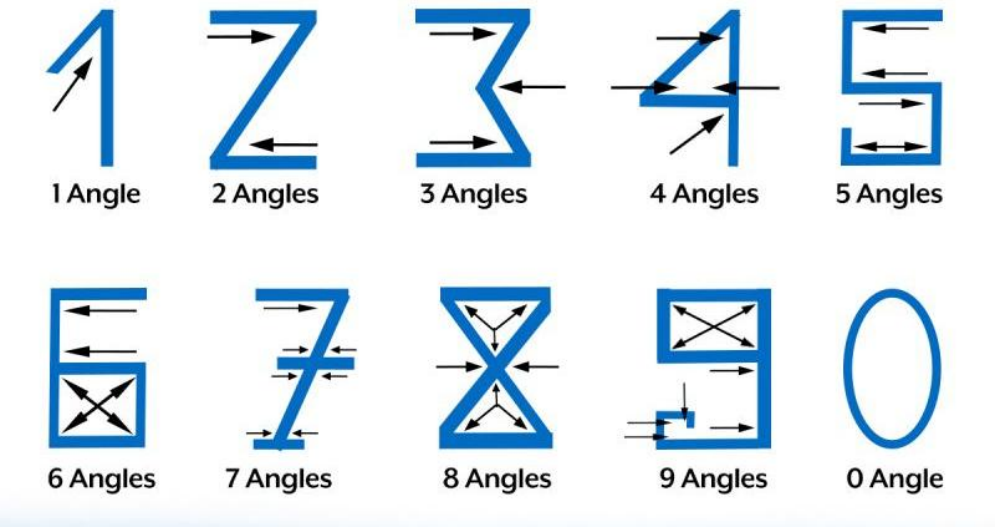
- This chapter explores how numerical ideas developed in the Vedic age.

Origins of Number Use

- Numbers first appear in the Rigveda in poetic and symbolic contexts.

- They were used to describe quantities of offerings, ritual objects, altar construction and cosmology

- Enumeration helped structure hymns and ritual procedures.

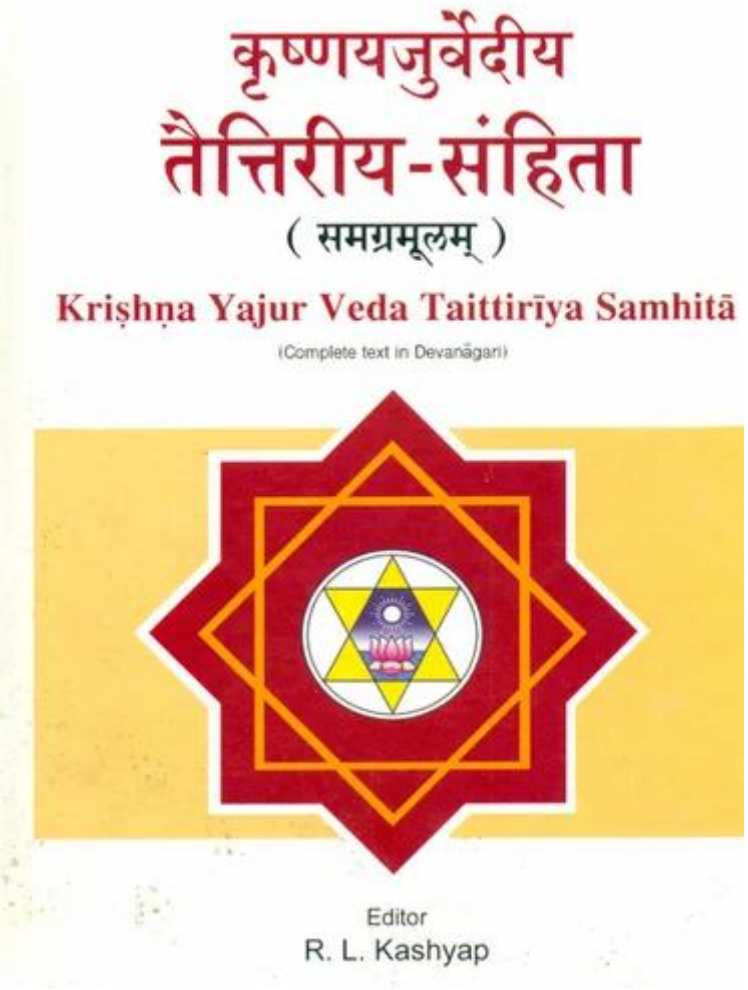


- Early number names reveal a strong decimal tendency.

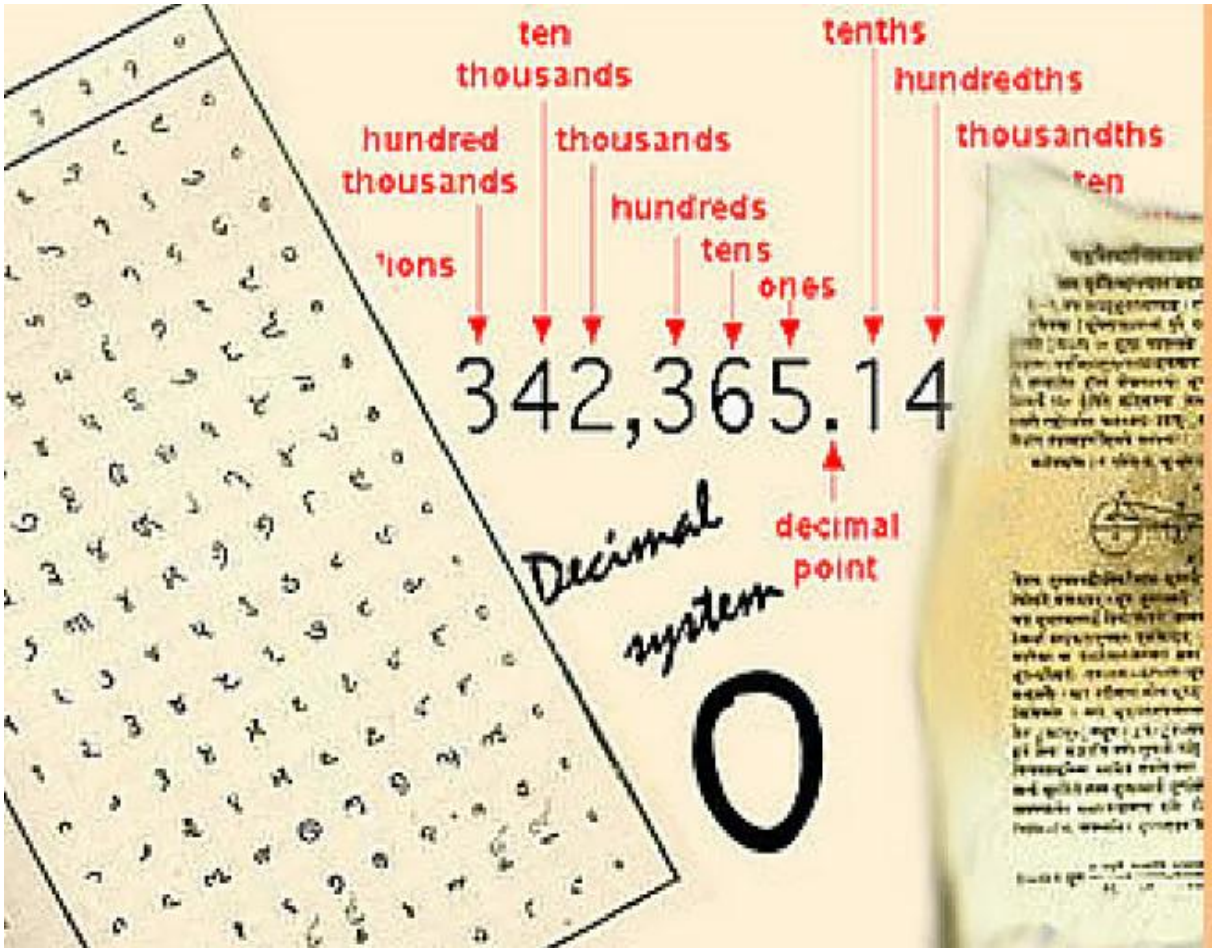
- These origins laid the foundation for later Indian arithmetic.

Number Names in Early Texts

- Distinct names existed for numbers from one to ten.
- Multiples of ten also had specific linguistic forms.
- The structure of number words shows awareness of grouping by tens.
- This linguistic pattern hints at the decimal system's roots.
- Language played a major role in shaping numerical thought.



Decimal Structure Emerging

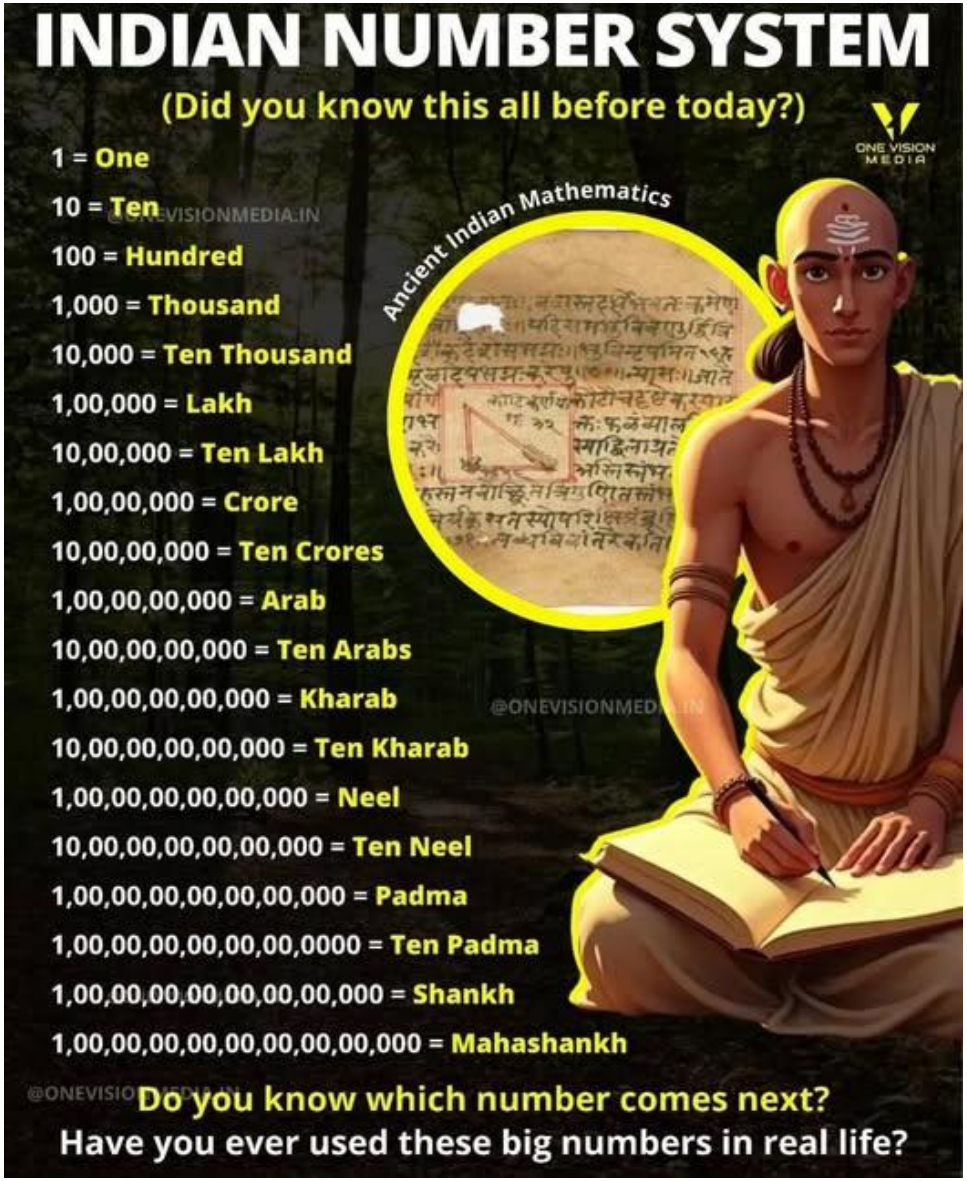


Decimal Structure Emerging

- Vedic number names indicate grouping in powers of ten.
- This shows early understanding of place value concepts.
- Though symbols were not yet used, the idea of base 10 was present.
- Large numbers were formed systematically using prefixes.
- This is a major step toward the Indian decimal system.

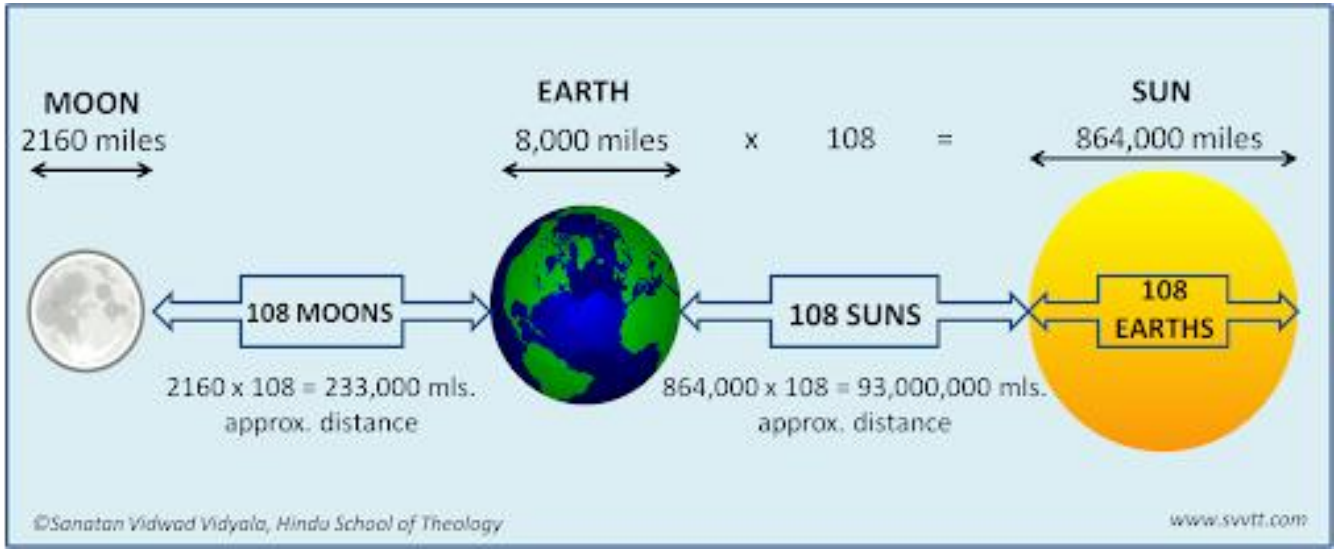
Large Numbers in the Vedas

- Texts mention numbers extending to extremely high magnitudes.
- These numbers often had ritual or symbolic meanings.
- The concept of very large numbers fascinated Vedic thinkers.
- It reflects comfort with numerical abstraction.
- Such thinking later influenced Indian mathematical traditions.



Symbolic Meaning of Numbers

- Numbers often had cosmological significance.
- Three represented the three worlds.
- Seven symbolized completeness in many hymns.
- Twelve was linked to the solar year.
- Thus numbers were both quantitative and symbolic.



Counting in Ritual Practice

- Precise counting was necessary for Vedic sacrifices.
- The number of bricks and altar layers was specified.
- Ritual manuals demanded numerical accuracy.
- This encouraged careful arithmetic habits.
- Practical needs supported mathematical growth.

Counting in Ritual Practice

Later Vedic Society:

Core Texts

Samhitas, Brahmanas

Archaeological Evidence

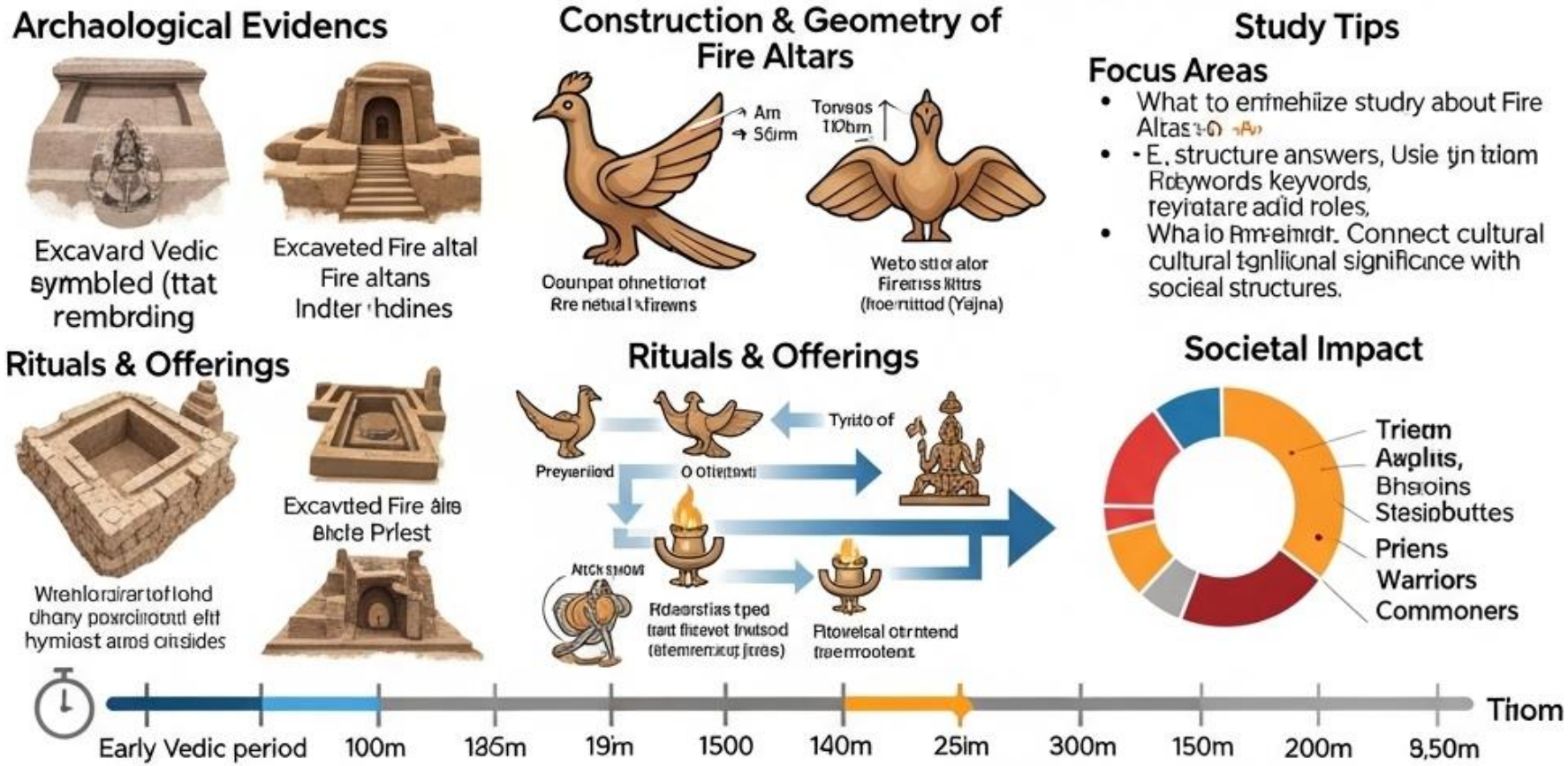
Mentinn Fire altans

Theological Significance

Wire altons symblolized cosmurication with deities perre

- Fire altons hymer (Rta) cosmic orden (Rta) and communications, with sociad deities

Counting in Ritual Practice



Number Sequences

- Sequences of numbers appear in liturgical recitations.
- They sometimes increase in regular patterns.
- These may reflect early combinatorial thinking.
- Ordered sequences helped memory and oral transmission.

$$N_n = N_{n-1} + N_{n-3} \text{ for } n > 2,\ N_0 = N_1 = N_2 = 1$$

- Pattern recognition was important in Vedic culture.

Arithmetic Sequence	Geometric Sequence	Triangular Sequence
$x_n = x1 + (n-1)\,d$	$x_n = x1 \times r^{n-1}$	$x_n = (n^2 + n) \div 2$

Cube Sequence	Fibonacci Sequence	Square Sequence
$x_n = n^3$	$a_{n+2} = a_{n+1} + a_n$	$x_n = n^2$

Infinity in Vedic Thought

- The idea of the infinite appears in philosophical passages.
- The famous phrase ‘purnam adah purnam idam’ suggests completeness.
- Numbers were seen as extendable without bound.
- This philosophical infinity influenced later mathematics.
- It laid groundwork for comfort with large numbers.

Re संस्कृत®

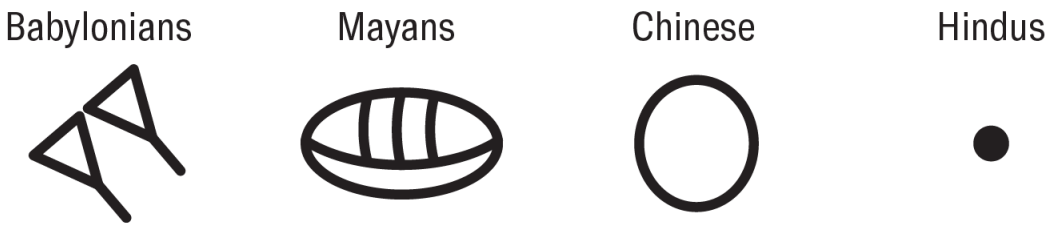
पूर्णमदः पूर्णमिदं पूर्णात् पूर्णमुदच्यते।
पूर्णस्य पूर्णमादाय पूर्णमेवावशिष्यते॥

Ishavasya Upanishad, Invocatory Verse | resanskrit.com

ॐ, वह (अज्ञात ब्रह्मांड) पूर्ण है, यह (प्रतीत ब्रह्मांड) पूर्ण है;
यह पूर्णता उस पूर्णता से प्राप्त हुई है;
जब यह पूर्णता उस परिपूर्णता से अलग हो जाती है,
तो जो भी रहता है वह पूर्णता है।

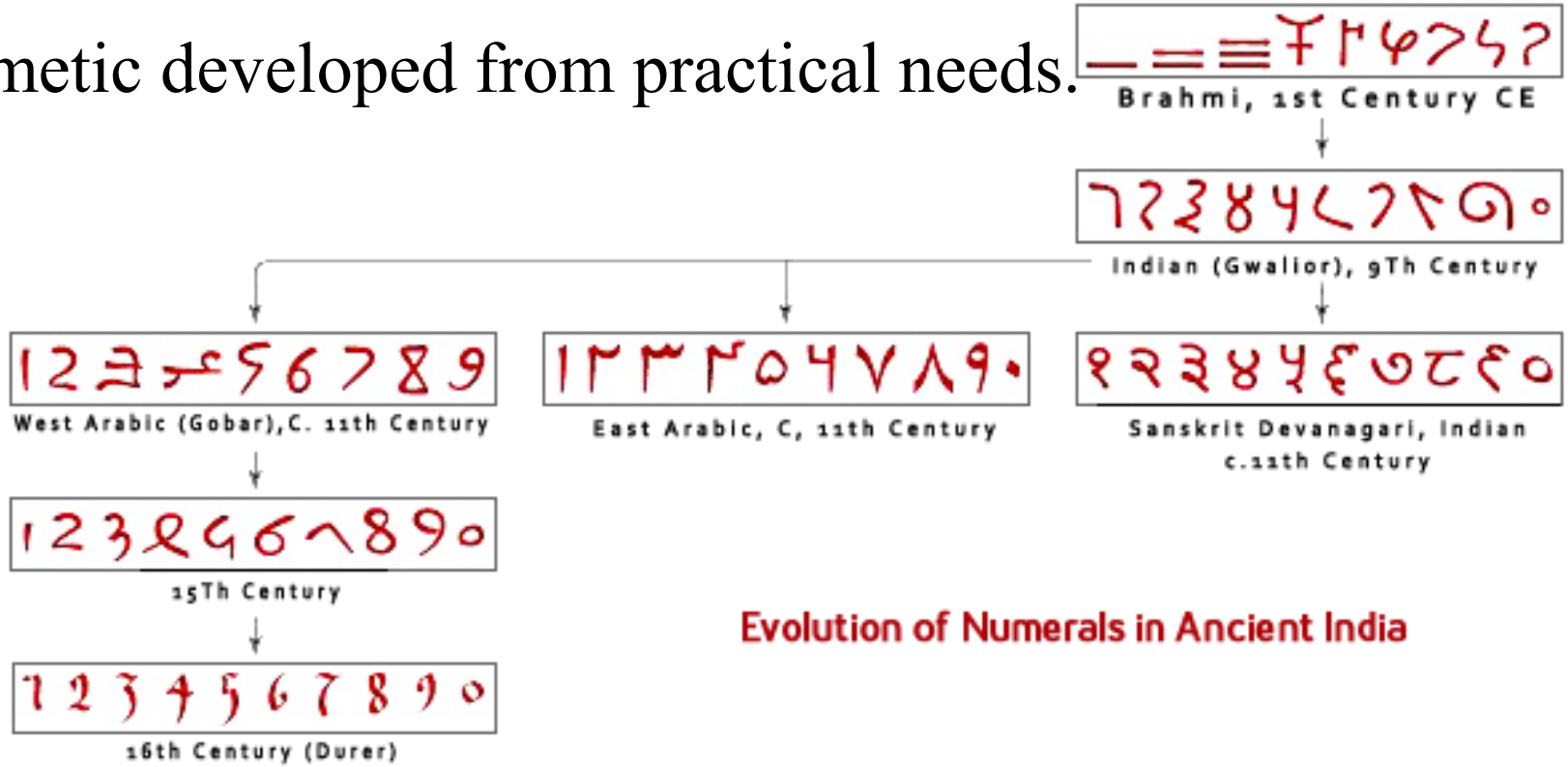
Concept of Zero (Early Notions)

- Explicit zero was not yet formalized in Vedic times.
- However, ideas of emptiness appear in philosophical texts.
- Absence and void had metaphysical meaning.
- These ideas may have prepared ground for zero later.
- Mathematical zero would emerge centuries afterward.



Arithmetic in the Vedic Period

- Basic operations like addition and subtraction were known.
- Multiplication appeared in the form of repeated addition.
- Division concepts were implied in sharing rituals.
- Fractions were used in ritual measurements.
- Arithmetic developed from practical needs.



Evolution of Numerals in Ancient India

Fractions and Parts

- Texts describe parts of offerings and measures.
- This implies familiarity with fractional quantities.
- Ritual accuracy required division of materials.
- Though not formalized, fraction thinking was present.
- Later mathematics would systematize fractions.
- However Indian mathematics never adopted the decimal representation of fractions throughout the course of its history

Combinatorial Ideas

- Prosody and hymn structures involved counting patterns.
- Meters depended on arrangements of syllables.
- This led to early combinatorial reasoning.
- Later scholars like Pingala (300 – 200 BCE) developed this further.

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

- Thus poetry influenced mathematical development.

Memory and Oral Transmission

- Vedic knowledge was preserved orally for centuries.
- Numbers helped structure memory devices.
- Regular numerical patterns aided recall.
- Mathematics and memory techniques were linked.
- This influenced the style of later Indian mathematics.

Numbers and Cosmology

- Numbers described cosmic cycles and time spans.
- Huge durations like Yugas were expressed numerically.
- This required handling very large numbers.
- Cosmology stimulated numerical imagination.
- Mathematics grew alongside cosmological speculation.

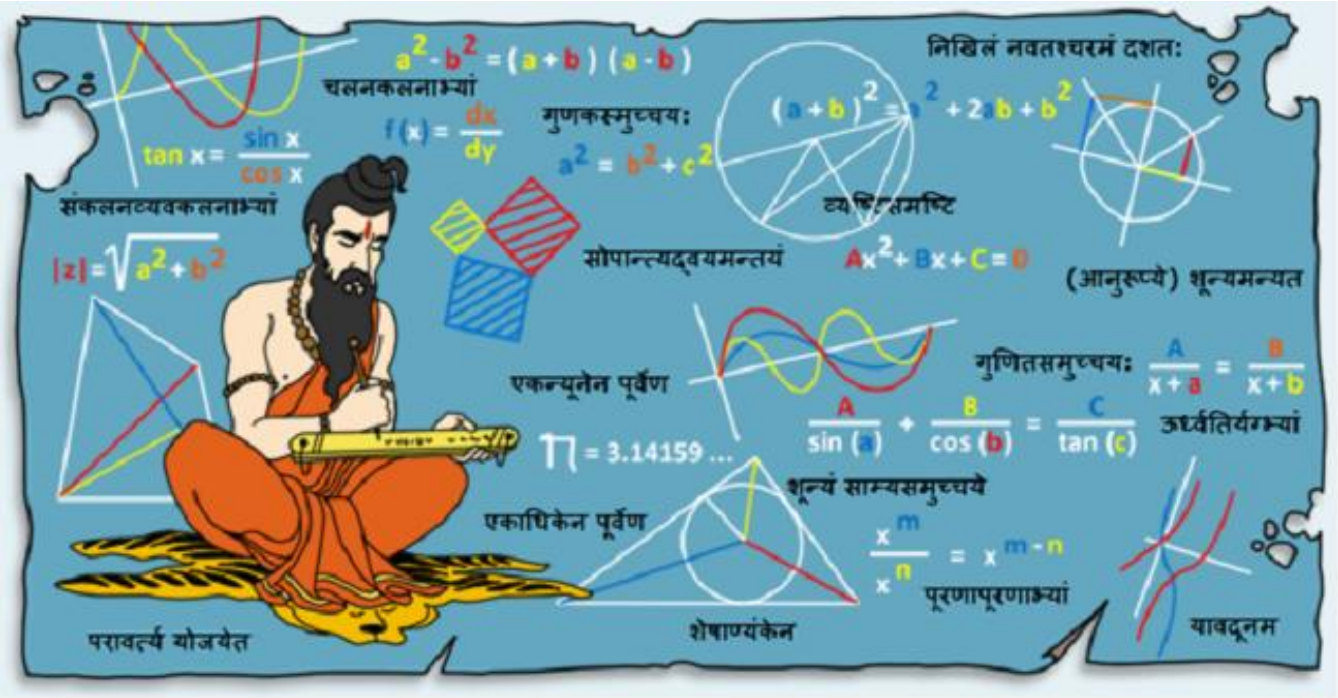
Powers of Ten

- Lists of powers of ten appear in later Vedic texts.
- Names existed for numbers up to extremely high powers.
- This demonstrates systematic numerical scaling.
- It reflects early place-value awareness.
- Such lists are unique in ancient cultures.

Powers of Ten in Ancient Indian Mathematics			
daśa	*daśa*	Ten	10^1
daśa daśa	*śata*	Hundred	10^2
daśa śata	*sahasra*	Thousand	10^3
daśa sahasra	*ayuta*	Ten Thousand	10^4
daśa ayuta	*niyuta*	Hundred Thousand	10^5
daśa niyuta	*prayuta*	Million	10^6
daśa prayuta	*arbuda*	Ten Million	10^7
daśa arbuda	*nyarbuda*	Hundred Million	10^8
daśa nyarbuda	*samudra*	Thousand Million	10^9
daśa samudra	*madhya*	Ten Thousand Million	10^{10}
daśa madhya	*anta*	Hundred Million Thousand	10^{11}
daśa anta	*parārdha*	Billion	10^{12}

Cultural Importance of Enumeration

- Enumeration was not merely practical but sacred.
- Counting was tied to order and harmony.
- Numbers represented structure in the universe.
- This gave mathematics a cultural prestige.
- It encouraged preservation of numerical knowledge.



Transition Toward Formal Mathematics

- Late Vedic texts show increasing numerical sophistication.
- Geometry appears in altar construction rules.
- Arithmetic grows more precise.
- Symbolism begins shifting toward calculation.
- This prepares the way for the classical period.

Comparison with Other Civilizations

- Vedic number usage differs from Babylonian and Greek traditions.
- Indian thought favored decimal grouping early on.
- Large numbers were more freely used in India.
- Symbolic and ritual contexts shaped usage.
- These differences influenced later mathematical styles.

Influence on Later Indian Mathematics

- Decimal thinking became central in later arithmetic.
- Comfort with large numbers led to advanced place value.
- Ritual precision evolved into algorithmic thinking.
- Philosophical infinity influenced calculus ideas centuries later.
- Thus Vedic ideas had long-term mathematical impact.

Language and Number Structure

- Sanskrit grammar influenced numerical terminology.
- Compound words made large numbers easier to express.
- Linguistic precision supported mathematical clarity.
- Terminology helped preserve numerical knowledge.
- Language was a key tool in mathematical evolution.

Educational Aspects

- Learning numbers was part of Vedic education.
- Students memorized number lists and sequences.
- Recitation reinforced numerical familiarity.
- This oral pedagogy shaped mathematical style.
- It emphasized rules and procedures over symbols.

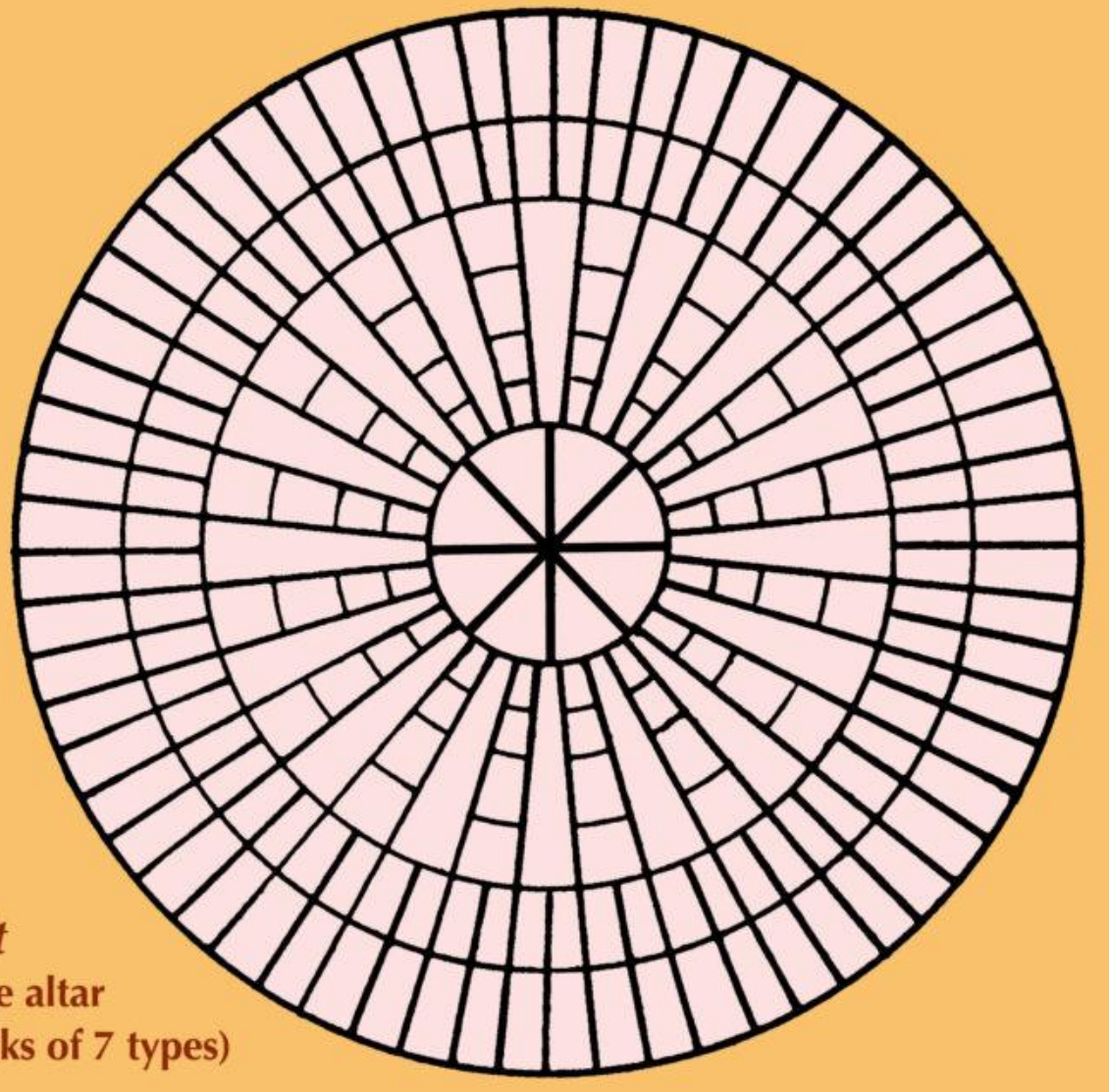
Limitations of Vedic Mathematics

- Symbolic notation was not yet developed.
- Proofs were not presented in formal structure.
- Mathematics remained tied to ritual contexts.
- Abstract algebra had not yet emerged.
- Future periods would expand these foundations.

Summary of Numbers in the Vedic Literature

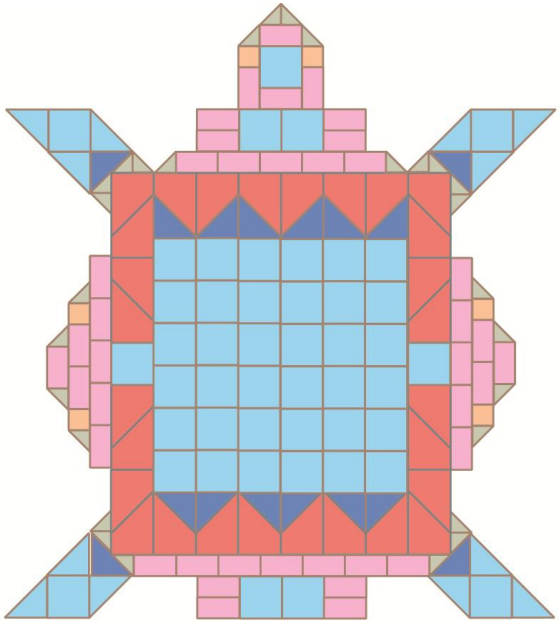
- Vedic literature reveals early numerical sophistication.
- Decimal structure was already forming.
- Large numbers and infinity were conceptually important.
- Arithmetic developed from ritual and cultural needs.
- These ideas became the roots of Indian mathematics.

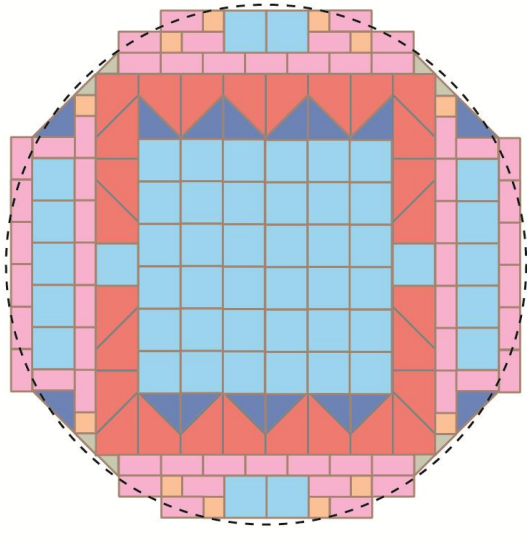
Appendices

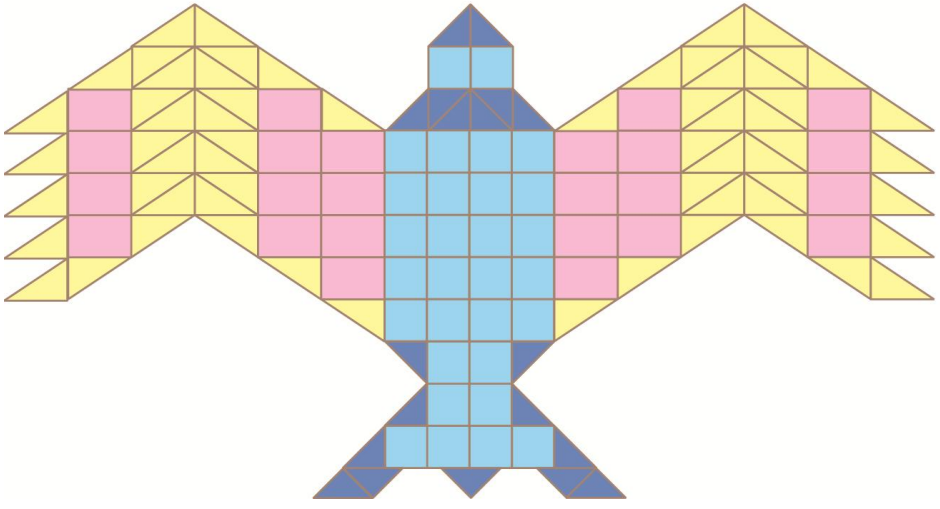


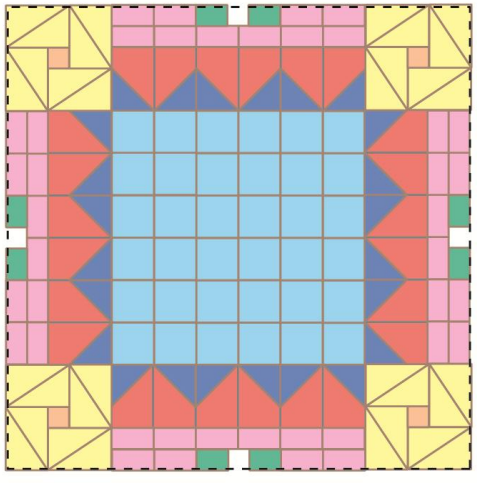
The *rathacakracit*
or chariot-wheel fire altar
(first layer, 200 bricks of 7 types)

Altar Variations









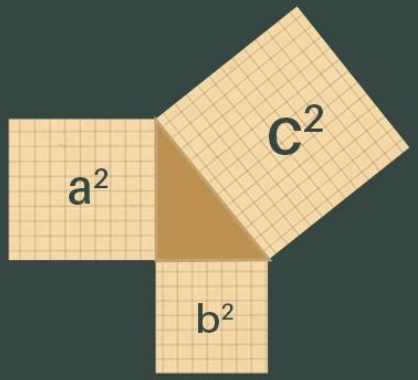
DID YOU KNOW : Pythagoras theorem is mentioned in Yajur Veda

Baudhayana Sulbasutra

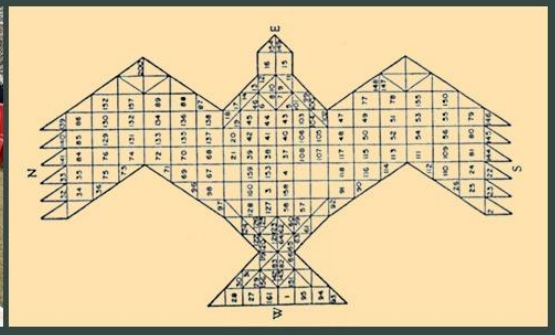
दीर्घचतुरश्रस्याक्ष्णया रज्जुः पार्श्वमानी तिर्यग्
मानी च यत् पृथग् भूते कुरूतस्तदुभयं करोति ॥

Baudhāyana used a rope as an example in the above shloka/verse, which can be translated as:

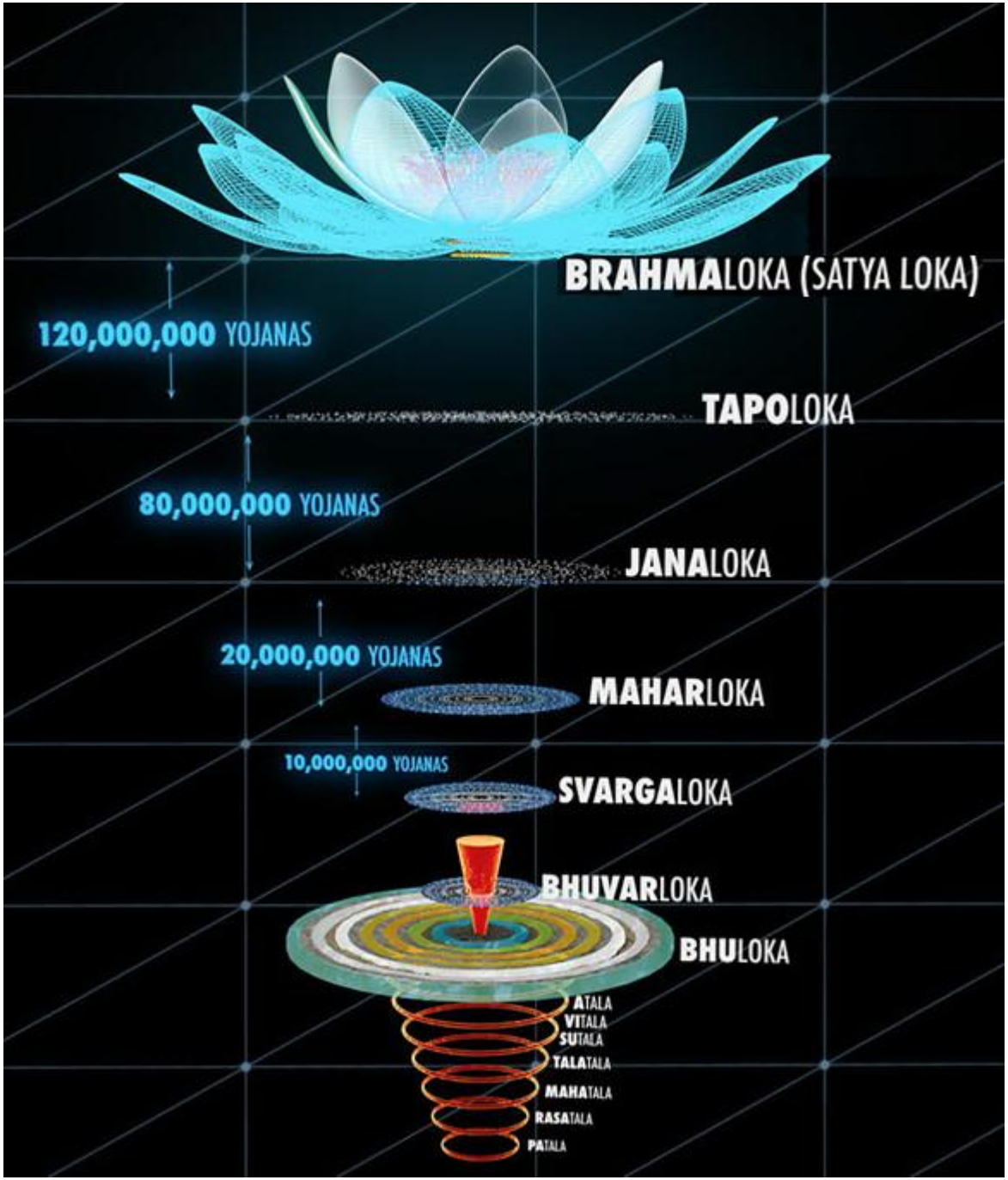
"A rope stretched along the length of the diagonal produces an area which the vertical and horizontal sides makes together"



$$c^2 = a^2 + b^2$$



The Sulbasutras is like a guide to the Vedas which formulate rules for constructing Vedi (altars). They contain valuable information about the geometry of ancient India, including instructions for constructing sacrificial altars (vedi) and locating sacred fires (agni). This formula is used in constructing Yagna Vedi (altars) in the shape of an eagle with Triangular tiles in a special type of Yagna called Agnicayana (a twelve day Yajna ceremony)


BRAHMALOKA (SATYA LOKA)
120,000,000 YOJANAS
TAPOLOKA
80,000,000 YOJANAS
JANALOKA
20,000,000 YOJANAS
MAHARLOKA
10,000,000 YOJANAS
SVARGALOKA
BHUVARLOKA
BHULOKA
ATALA
VITALA
SUTALA
TALATALA
MAHATALA
RASATALA
PATALA

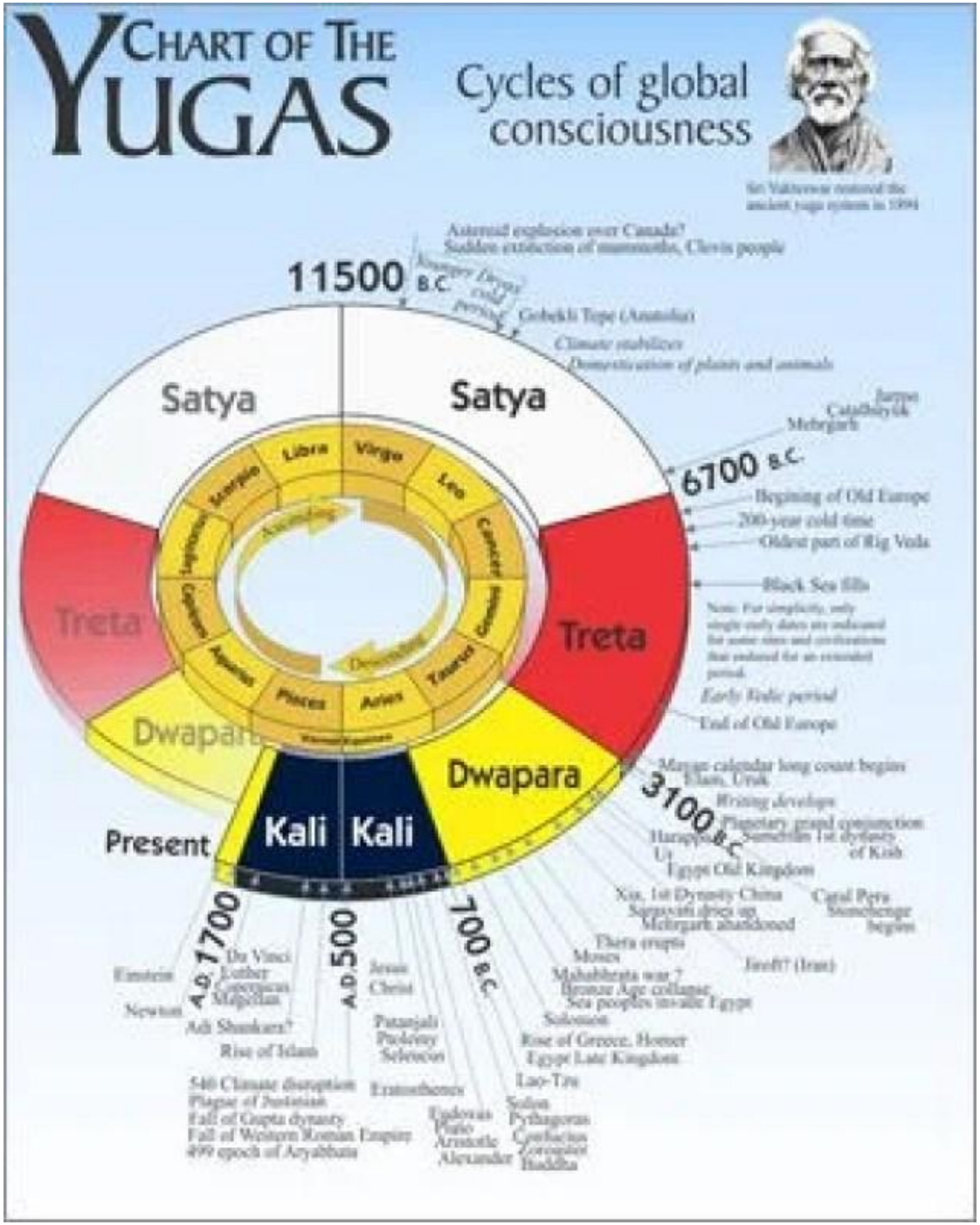
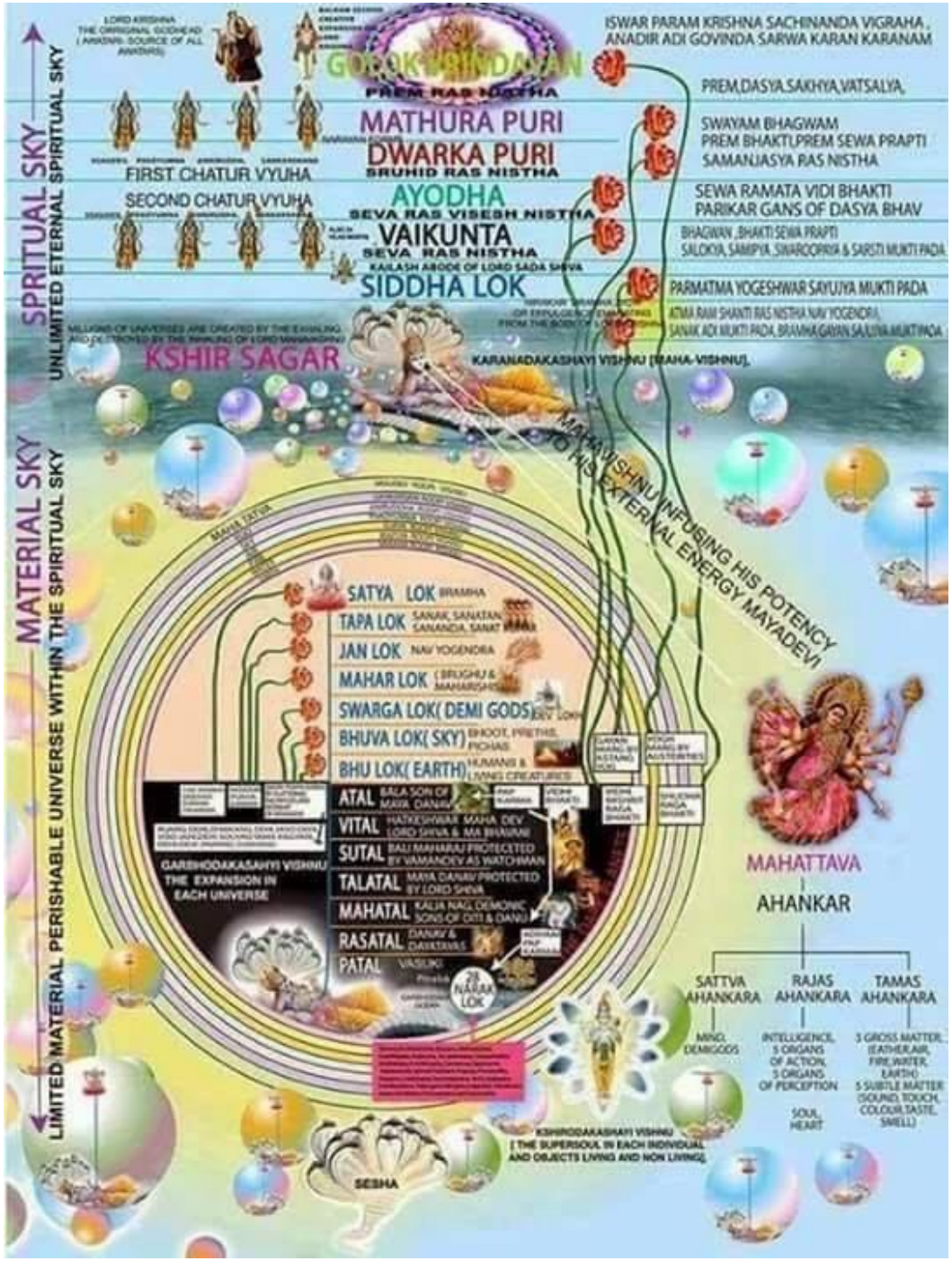
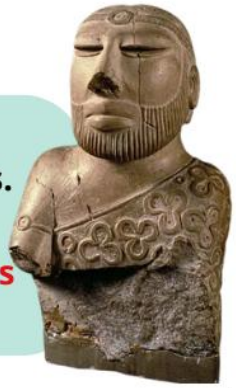


CHART OF THE YUGAS
Cycles of global consciousness
Sri Yukteswar restored the ancient yuga system in 1894
Asteroid explosion over Canada?
Sudden extinction of mammoths, Clovis people
11500 B.C.
Younger Dryas cold period
Gobekli Tepe (Anatolia)
Climate stabilizes
Domestication of plants and animals
Satya
Satya
Jericho
Catalhoyuk
Mehrgarh
6700 B.C.
Begining of Old Europe
200-year cold time
Oldest part of Rig Veda
Black Sea fills
Note: For simplicity, only single early dates are indicated for some sites and civilizations that endured for an extended period.
Early Vedic period
End of Old Europe
Libra
Virgo
Leo
Cancer
Gemini
Taurus
Aries
Pisces
Aquarius
Capricorn
Sagittarius
Scorpio
Ascending
Descending
Treta
Treta
Dwapara
Dwapara
Present
Kali
Kali
Mayan calendar long count begins
Elam, Uruk
Writing develops
Planetary grand conjunction
Harappa
Sumerian 1st dynasty of Kish
Ur
Egypt Old Kingdom
3100 B.C.
Xia, 1st Dynasty China
Sarasvati dries up
Mehrgarh abandoned
Canal Peru
Stonehenge begins
Thera erupts
Moses
Mahabharata war ?
Bronze Age collapse
Sea peoples invade Egypt
Jiroft? (Iran)
Solomon
Rise of Greece, Homer
Egypt Late Kingdom
Lao-Tze
Solon
Pythagoras
Confucius
Zoroaster
Buddha
700 B.C.
Jesus Christ
Patanjali
Ptolemy
Seleucus
Eratosthenes
Eudoxus
Plato
Aristotle
Alexander
A.D. 500
540 Climate disruption
Plague of Justinian
Fall of Gupta dynasty
Fall of Western Roman Empire
499 epoch of Aryabhata
A.D. 1700
Da Vinci
Luther
Copernicus
Magellan
Einstein
Newton
Adi Shankara?
Rise of Islam


SPRITUAL SKY
UNLIMITED ETERNAL SPIRITUAL SKY
MATERIAL SKY
LIMITED MATERIAL PERISHABLE UNIVERSE WITHIN THE SPIRITUAL SKY
LORD KRISHNA
THE ORIGINAL GODHEAD
(ANADI: SOURCE OF ALL AVATARS)
GOLOK VRINDAVAN
PREM RAS NISTHA
MATHURA PURI
DWARKA PURI
SRUHID RAS NISTHA
FIRST CHATUR VYUHA
SECOND CHATUR VYUHA
AYODHA
SEVA RAS VISESH NISTHA
VAIKUNTA
SEVA RAS NISTHA
KAILASH ABODE OF LORD SADA SIVA
SIDDHA LOK
KSHIR SAGAR
KARANADAKASHAYI VISHNU [MAHA-VISHNU],
ISWAR PARAM KRISHNA SACHINANDA VIGRAHA ,
ANADIR ADI GOVINDA SARWA KARAN KARANAM
PREM,DASYA,SAKHYA,VATSALYA,
SWAYAM BHAGWAM
PREM BHAKTI,PREM SEWA PRAPTI
SAMANJASYA RAS NISTHA
SEWA RAMATA VIDI BHAKTI
PARIKAR GANS OF DASYA BHAV
BHAGWAN ,BHAKTI SEWA PRAPTI
SALOKYA, SAMIPYA ,SWAROOPRIR & SARSTI MUKTI PADA
PARMATMA YOGESHWAR SAYUJYA MUKTI PADA
ATMA RAM SHANTI RAS NISTHA NAV YOGENDRA,
SANAK ADI MUKTI PADA, BRAHMA GAYAN SALUNA MUKTI PADA
MAHAVISHNU INFUSING HIS POTENCY
TO HIS EXTERNAL ENERGY MAYADEVI
SATYA LOK BRAMHA
TAPA LOK SANAK, SANATAN, SANANDA, SANAT
JAN LOK NAV YOGENDRA
MAHAR LOK (BRUGHU & MAHARISHI)
SWARGA LOK(DEMI GODS) DEV LOKH
BHUVA LOK(SKY) BHOOT, PRETHS, PICHAS
BHU LOK(EARTH) HUMANS & LIVING CREATURES
ATAL BALA SON OF MAYA DANAV
VITAL HATKESHWAR MAHA DEV LORD SHIVA & MA BHAVANI
SUTAL BALI MAHARAJ PROTECETED BY VAMANDEV AS WATCHMAN
TALATAL MAYA DANAV PROTECTED BY LORD SHIVA
MAHATAL RAJA NAG, DEMONIC SONS OF DITI & DANU
RASATAL DANAV & DAVATAVAS
PATAL VASUKI
GARBHODAKASAHYI VISHNU
THE EXPANSION IN
EACH UNIVERSE
MAHATTAVA
AHANKAR
SATTVA AHANKARA
RAJAS AHANKARA
TAMAS AHANKARA
MIND, DEMIGODS
INTELLIGENCE, 5 ORGANS OF ACTION, 5 ORGANS OF PERCEPTION
SOUL, HEART
5 GROSS MATTER, (EATHER,AIR, FIRE,WATER, EARTH) 5 SUBTLE MATTER (SOUND, TOUCH, COLOUR,TASTE, SMELL)
KSHIRODAKASHAYI VISHNU
[THE SUPERSOUL IN EACH INDIVIDUAL AND OBJECTS LIVING AND NON LIVING],
SESHA

Indus Valley Civilization



Indus Valley Civilisation differs from other ancient civilizations because inequality levels were far lower than in other civilizations. This was because the Indus, unlike the other rivers, changed its course often. This changed the structure of the society of the Indus Valley Civilization

Civilization	Indus Valley	Nile	Mesopotamia	Yellow River
Period	2600-1900 BCE	3100-1070 BCE	4000-2500 BCE	5000-3000 BCE
Location	South Asia (Pakistan and northwest India)	Northeast Africa (Egypt)	Middle East (Iraq and surrounding areas)	East Asia (China)
Key Cities	Harappa, Mohenjo-Daro	Memphis, Thebes	Ur, Uruk, Babylon	Anyang
Writing System	Undeciphered script	Hieroglyphics, hieratic script	Cuneiform	Oracle bone script
Religion	Polytheistic (mother goddess)	Polytheistic (emphasis on pharaoh as divine)	Polytheistic (emphasis on city-state patron deities)	Ancestor worship, with some evidence of **shamanism** and **totemism**
Architecture	Large, planned cities with brick buildings, public baths, and sewage systems	**Pyramids**, temples, and monumental tombs	**Ziggurats**, palaces, and monumental gates	**Large walled cities with** ceremonial architecture and burials of elite
Trade	Likely traded with Mesopotamia and Gulf region for luxury goods	Traded with Nubia, Levant, and Mesopotamia for luxury goods	Traded with Indus Valley and Persian Gulf region for luxury goods	Traded with Southeast Asia, Korea, and Japan for luxury goods
Decline	Unknown, potentially due to environmental factors or invasion	Conquered by foreign invaders and assimilated into larger empires	Declined due to political instability and invasion	Conquered and assimilated into larger empires

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